Tutorial -3

Q1: Convert the following regular expressions to NFAs using the procedure to prove that NFAs are equivalent to Res. In all parts E = {a, b}.

a) a(abb) * U b
b) a+ U (ab)+
c) (a U b+)a+b+

Q2: Use the pumping lemma to show that the following languages are not regular.

a) A= {www | w in {a,b}*}
b) A={a^{2n}| n >0}

Q3:

Describe the error in the following "proof" that 0* 1* is not a regular language. The proof is by contradiction. Assume that 0* 1* is regular. Let p be the pumping length for 0* 1* given by the pumping lemma. Choose s to be the string O^p1^p, which cannot be pumped. Thus you have a contradiction. So 0* 1* is not regular.

Q5: Show that the follwing is not regular : L = {a^K | k is a prime number}

Q6: Show that the follwing is not regular : L = {a^n b^{n+1}, n>0 }

Q7: Show that the follwing is not regular : L = {a^n b^{2n}, n>0 }

Q8: Show that the follwing is not regular : TRAILING-COUNT as any string s followed by a number of a's equal to the length of s.

Q9: Show that the follwing is not regular EVENPALINDROME = { all words in PALINDROME that have even length}

Q10: L = { w w^R | w is {a, b}* }; R implies reverse